

Bikram Pandit

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Professional Summary

PhD candidate specializing in multi-robot control, reinforcement learning, and sim-to-real transfer, with 5 publications at top peer-reviewed robotics conferences (ICRA, CoRL), including an Oral and a Spotlight, a 1st-place finish in the DARPA TIAMAT Challenge, and policies deployed on 6+ real robot platforms spanning bipeds, quadrupeds, and humanoids.

Education

Oregon State University <i>PhD in Artificial Intelligence</i> Research focus: large-scale robot learning, decentralized control, sim-to-real transfer, and loco-manipulation systems	Corvallis, Oregon <i>Sept 2021 – Jan 2027 (Expected)</i>
Rungta College of Engineering and Technology <i>Bachelor's in Computer Science and Engineering</i>	India <i>Aug 2013 – June 2017</i>

Awards and Honors

- **1st Place – DARPA TIAMAT Challenge** **2026**
Developed robust Spot policies for high-speed exploration, fall recovery on unpredictable terrain, and obstacle avoidance in cluttered settings; integrated with an autonomous planner for sequential task execution.
- **Best Paper Finalist** **ICRA, 2024**
Learning Vision-Based Bipedal Locomotion for Challenging Terrain.
- **1st Prize, Lightning Talk – OSU AI Meetup for Industry and Alumni** **2024**
Recognized for presenting “Learning Vision-Based Bipedal Locomotion for Challenging Terrain.”
- **2nd Place – AgAID Institute Hackathon** **2023**
Competed against 32 teams from 8+ universities in a Microsoft-sponsored hackathon.

Research Experience

Dynamic Robotics and AI Lab, Oregon State University <i>Research Assistant</i>	Corvallis, Oregon <i>Sept 2021 – Present</i>
○ Led development of scalable reinforcement learning systems for bipedal, humanoid, quadruped-with-arms, and multi-robot platforms. ○ Designed vision-based RL policies integrating proprioception and onboard camera inputs for terrain-aware locomotion and manipulation. ○ Developed decentralized and multi-agent control policies enabling cooperative manipulation and payload transport. ○ Built distributed PPO, IPPO, and MAPPO training pipelines in IsaacLab and MuJoCo supporting large-scale parallel simulation. ○ Implemented sim-to-real transfer using domain randomization, sensor noise modeling, and robustness training. ○ Deployed learned policies on real robots, including Cassie, Digit, Spot, Unitree Go2, and H1, using accelerated inference frameworks such as NVIDIA TensorRT.	
RAI Institute (formerly Boston Dynamics AI Institute) <i>Robotics Research Intern</i>	Cambridge, Massachusetts <i>Sept 2024 – Jan 2025</i>

- Collaborated with research teams on humanoid locomotion and loco-manipulation systems.
- Developed multi-instance Isaac Sim infrastructure supporting heterogeneous robot articulations for scalable data generation.
- Built unified imitation learning pipelines to consolidate multiple specialized controllers into a single deployable policy.
- Validated natural walking and running behaviors in simulation and on real humanoid hardware.

Publications

- **Learning Multi-Humanoid Pickup and Transport via Decentralized Object-Centric Control** Under review
Object-centric control abstraction unifying single- and multi-robot humanoid pickup, transport, and handoff, validated on Digit V3 humanoids.
- **Multi-Quadruped Cooperative Object Transport: Learning Decentralized Pinch-Lift-Move** ICRA, 2026
Gripperless pinch-lift-move coordination for quadruped-arm teams of 2–10, validated on Unitree Go2 quadrupeds with D1 arms.
Project Page · Code
- **Non-conflicting Energy Minimization in RL Based Robot Control (Oral)** CoRL, 2025
Hyperparameter-free energy optimization for RL control with sim-to-real validation.
Project Page · Code
- **Learning Decentralized Multi-Biped Control for Payload Transport (Spotlight)** CoRL, 2024
Decentralized policy scaling rigid cooperative carrying to arbitrarily sized teams of bipeds sharing one carrier, validated on Cassie robots.
Project Page · Code
- **Learning Vision-Based Bipedal Locomotion for Challenging Terrain (Best Paper Finalist)** ICRA, 2024
Vision-conditioned sim-to-real locomotion robust to challenging terrain without explicit state estimation.
arXiv
- **Simple Masked Training Strategies Yield Control Policies That Are Robust to Sensor Failure** CoRL, 2024
Robust policy training improving resilience to sensor dropout in real-world deployments.
Paper

Patents

- **Systems and Methods for Decentralized Multi-Robot Cooperative Transport** Under review

Industry Experience

MiCROTEC Inc.

Machine Learning Intern

June 2023 – Aug 2023

- Accelerated large-scale vision model training by 80% through optimized data pipelines and system-level performance tuning.
- Developed robust segmentation models addressing class imbalance and deployment constraints for production systems.
- Integrated model training, evaluation, and deployment into automated ML DevOps pipelines using Docker, CI, and experiment tracking.

Digital Product Labs

Senior Software Developer

Dec 2017 – June 2021

- Designed and deployed production machine learning systems across mobile, web, and cloud platforms.
- Integrated machine learning components into consumer-facing products, improving performance and scalability in production.
- Partnered with product and engineering teams to scope and ship AI-driven features end to end.